

Richard Pillaca

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Education

The University of British Columbia

Sept 2021 – Dec 2025

Bachelor of Science in Computer Science, Minor in Economics

Kelowna, Canada

- **Certificates:** Power BI Data Analyst (**Microsoft**), Scrum Master (**Scrum.org**), Claude Certified Architect (**Anthropic**)
- **Relevant Coursework:** Software Engineering, Data Analysis, Data Structures & Algorithms, UX/UI Design, Web Development

Work Experience

BrainTrainr

Jan 2026 – May 2026

Web Content & UX Lead | [LinkedIn](#)

Toronto, Canada

- Led **UI/UX analysis** and design implementation in **Figma**, then collaborated with the team to build and ship **accessible** interfaces for the **Universal Voice AI Platform** launch (BrainTrainr 1800).
- Turned approved **Figma** designs into a **responsive WordPress** site, removing design-to-development friction to keep delivery on schedule.

Karac

Oct 2025 – Feb 2026

Frontend Developer | [Site](#)

Pully, Switzerland

- Cut average user signup time from 2 min to 45 seconds (**63% faster**) by redesigning the **onboarding flow**: verification code fields advance via **event-driven** logic as you type, and form errors appear instantly, no page reload required.
- Reduced **QA test environment setup** from 15 min to 30 seconds by building a one-click tool that fills the database with realistic test data, letting engineers **troubleshoot** session persistence and **secure data handling** immediately.
- Designed a modular authentication system with **reusable components** using **PHP** and **Laravel Socialite** to abstract provider logic, saving 5–10 hours of monthly maintenance while supporting **Google and Facebook sign-in**.

University of British Columbia

May 2025 – Aug 2025

Full-Stack Developer | [Case Study](#)

Kelowna, Canada

- Built a **scalable** Exam Analysis System using **Python**, **Django**, and **PostgreSQL** that processes **100k+** **student records** to automate grading, flag academic integrity issues, and deliver performance **analytics** and **reporting** for instructors.
- Reduced load times for complex reports from 4 min to 60 seconds by **optimizing** queries with server-side pagination and SQL aggregations, collaborating with a **cross-functional** team (Product, Design, QA) in an **Agile environment**.
- Developed a statistical algorithm to calculate student performance that reduced grading time from 4 hours to 15 minutes, verified using a rigorous **CI/CD** pipeline and **GitHub Actions** that ensured a 100% pass rate.

University of British Columbia

Jan 2025 – May 2025

Research Assistant | [Case Study](#) | [Paper](#)

Kelowna, Canada

- Engineered a **Python** and **GitHub API** pipeline to filter 5,000+ repositories, using statistical tests to evaluate AI vs. human code maintainability. AI code held its own in quality, but over-commented and over-engineered its solutions.
- Collaborated with a 4-person team to earn a nomination for the **Mining Software Repositories** Conference. Won “**Best Presentation**” out of 10 cohorts by translating complex statistical findings into clear, non-technical insights.

Projects

ScoutLane

 | [Case Study](#) | [Live](#) | *Next.js, TypeScript, PostgreSQL with Prisma ORM*

Apr 2026 – May 2026

- Architected an API-driven recruitment **web app** featuring a data-dense, job-scoped admin dashboard.
- Leveraged AI to asynchronously extract structured data from resumes, empowering teams to instantly filter, group, and analyze applicant metrics.

Bike-Share Network Optimization Analysis

 | [Case Study](#) | *R, Power BI, Excel, SQL*

Jan 2024 – Apr 2024

- Constructed an **ETL data pipeline** in **R** to process millions of historical records across large **datasets**, implementing regex-based cleaning logic to reduce noise by 15%. Designed **reporting** dashboards in **Power BI** and **Excel**, translating complex metrics into actionable strategies.

Skills

Languages & Data Layer: Java, Python, JavaScript/TypeScript, PHP, SQL, PostgreSQL, MySQL, MongoDB, Supabase

Frameworks & Libraries: React, Next.js, Node.js, Django, FastAPI, Laravel Socialite, Spring Boot, Alpine.js, Tailwind CSS

DevOps & Tools: Agile/Scrum, Git, CI/CD, AWS, Vercel, Docker, Cloudflare, REST APIs, GraphQL, Webhooks, Event-driven Architecture, Kafka, JUnit, PyTest, Playwright, Datadog, PostHog

Analytics & AI: ETL, Data Modeling, Pandas, Excel, Power BI, LLM Integration, RAG, AI Agents, LangGraph, MCP, Vercel AI